Data Structures and Algorithms 2

Question 4: Needleman–Wunsch Sequence Alignment

GitHub repository: https://github.com/janadatatech/dsa2-coursework

Coursework Question 4: Sequence alignment (20 marks)

No coding question. Please solve the following string alignment problem using Needleman–Wunsch. You need to use the optimal alignment algorithm with the strings **ATTACACG** and **CACAGAGT**.

Please provide the alignment matrix, the traceback and the string alignment using the scoring present in Unit 6.1.

What would happen if the penalty for a substitution is bigger than the penalty for a gap? Would that change the optimal string alignment? If so, what would it be? Please give only the new string alignment.

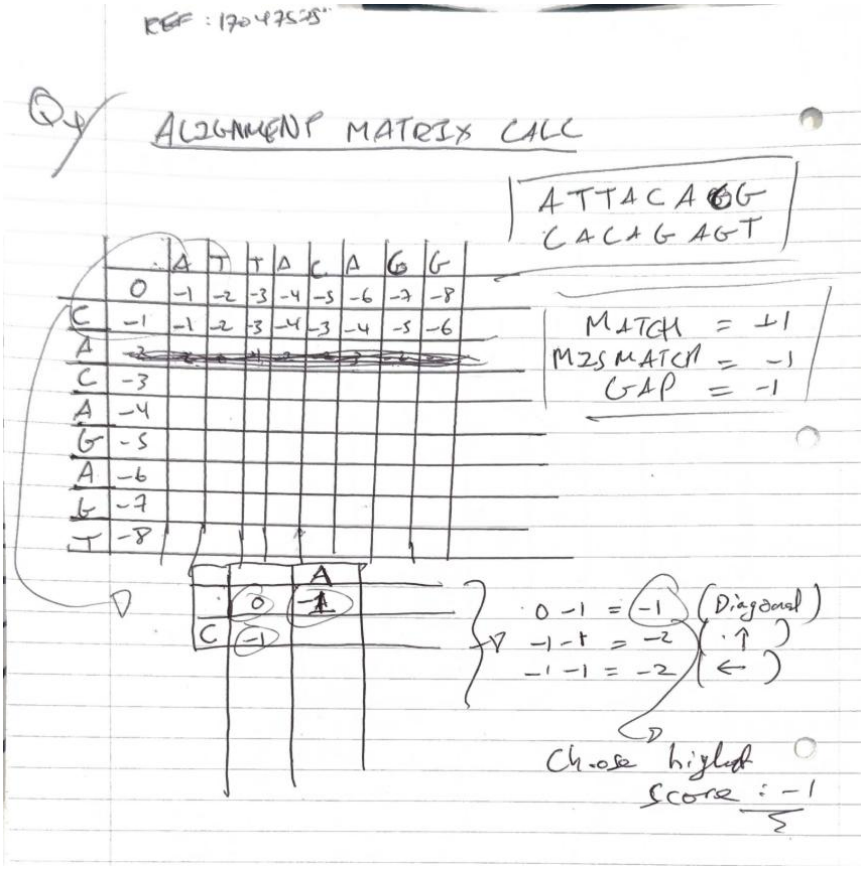
The Scoring rules

A match scores +1.
A mismatch scores −1.
A gap scores −1.

In this case I'm using dynamic programming where each cell uses scores already calculated in nearby cells and then I choose the highest score.

Alignment matrix

		A	T	T	A	C	A	C	G
	0	−1	−2	−3	−4	−5	−6	−7	−8
C	−1	−1	−2	−3	−4	−3	−4	−5	−6
A	−2	0	−1	−2	−2	−3	−2	−3	−4
C	−3	−1	−1	−2	−3	−1	−2	−1	−2
A	−4	−2	−2	−2	−1	−2	0	−1	−2
G	−5	−3	−3	−3	−2	−2	−1	−1	0
A	−6	−4	−4	−4	−2	−3	−1	−2	−1
G	−7	−5	−5	−5	−3	−3	−2	−2	−1
T	−8	−6	−4	−4	−4	−4	−3	−3	−2



My rough working shows how I calculated the first cell comparing C with A. I checked the diagonal, above and left scores, then chose the highest result which is −1. I used the same method to fill the remaining cells.

The Traceback

The starting scores decrease by 1 because each step represents adding a gap, so I start at the bottom-right cell and work backwards to the 0 in the top-left. The path shows which letters or gaps form the final alignment.

For reference the starting point is highlighted as blue which is the bottom-right cell, with a score of −2 and I follow the highlighted cells backwards to 0 highlighted in green.

My moves are up, diagonal, diagonal, up, diagonal, diagonal, diagonal, diagonal, left and left where each move tells me what to align. Moving diagonal aligns two letters, moving up puts a gap in the top sequence and moving left puts a gap in the bottom sequence.

This uses the scores already stored in the matrix, which is part of dynamic programming which is why I reverse the result because I followed the path backwards.

The Final alignment

A T T A C A – C G -
- - C A C A G A G T

A dash represents a gap therefore removing the gaps gives the two original sequences. As you can see there are 4 matches, 2 mismatches and 4 gaps which produces a total score of –2 :

Total score = 4 − 2 − 4 = −2.

This matches the bottom-right score in the matrix, so this shows that the alignment is optimal.

However that being said, there are other combinations route wise that are tied choices which can also give other optimal alignments with the same score. For example, when exploring a substitution penalty of −3, with a gap penalty of −1 and a match score of +1 you would get the following alignment:

A T T A C A C - - - G -
- - - - C A C A G A G T